

SHUBHANK GAUR

Lead Systems Architect & Engineering Manager

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(Open to Relocation & Visa Sponsorship)

PROFESSIONAL SUMMARY

Hands-on **Lead AI Systems Architect & Harness Engineer** with 12+ years of overall experience engineering distributed backend microservices, LLM execution engines, and internal developer SDKs. Proven track record designing asynchronous worker orchestrators in **Python**, implementing model routing and caching strategies for generative AI pipelines, and managing relational databases (**PostgreSQL**). Expert in **Agent Runtimes & State Machines, Python Backend Systems, SDK Architecture, Docker / Cloud Infrastructure**, and **Cost Optimization**.

CORE TECHNICAL SKILLS

- Agent Harness & Control:** Execution Engines, Graph/State Machine Runtimes, Context Management, Model Routing, Session Checkpoints, Guardrail/Policy Engines.
- Backend & Distributed Systems:** Python (5+ yrs production), Go, Node.js, REST APIs, WebSockets, Microservices, Async Queues (Redis, Pub/Sub, Kafka), C# / .Net, GraphQL, REST APIs.
- Databases & Memory Infrastructure:** PostgreSQL (Schema Design & Query Tuning), Redis Caching, Vector & Memory Concepts, MySQL, State Sync.
- SDK & Developer Tools:** Custom SDK Authoring, Internal Tooling & DX, Docker, AWS / GCP, GitHub Actions, CI/CD, Technical Specs (SPEC.md).

PROFESSIONAL EXPERIENCE

GamerzDan, Inc.	U.S.
Lead AI Systems Architect & Harness Engineer	<i>Dec 2018 – Present</i>
AI Agent Solutions & Infrastructure Division (MascotAI)	<i>Apr 2025 – Present</i>
Agent Execution Engine & Async Queues: Designed and implemented an asynchronous execution harness on AWS/GCP using Python, Docker, and Redis priority queues to orchestrate long-running execution loops for text-to-speech (Qwen-TTS) and 3D motion models (HY-Motion). Model Routing & Compute Optimization: Architected an on-demand GPU inference routing pipeline using rented compute clusters (Vast.ai), cutting infrastructure compute costs by 40%+ while maintaining high task completion success rates. Architecture Specs & Session Checkpoints: Authored technical reference specifications (SPEC.md) and session checkpoint recovery mechanisms to manage agent state, memory persistence, and system observability. FinTech OCR Data Pipeline: Built automated data processing harnesses using OCR, LLM prompt flows, and PostgreSQL to parse, categorize, and validate unstructured financial statements into clean database schemas.	
Games, Distributed Systems & Core Client Division	<i>Dec 2018 – Mar 2025</i>

- **High-Throughput State Synchronization:** Architected low-latency WebSocket backends and database state sync for live systems, including **NinjaFantasy** (\$300M peak market cap) and **PetLegends** (Solana ecosystem partner).
- **Production Software Releases:** Engineered gameplay, client logic, and backend interfaces for shipped consumer products including **Desktop Mascot Engine** (30k downloads, featured on Steam homepage) and **Run Zeus Run** (100k+ downloads).
- **Automated CI/CD Workflows:** Built **Python** automated build and testing pipelines with 1-click deployments across Steam, Android, and Discord preview build channels.

Transfivive Technet Pvt. Ltd.

Software Engineer & Engineering Manager

India

July 2012 – Oct 2018

- **SDK Authoring & Internal Tooling:** Regular code contributor and tool author for the popular **MMORPGKIT** open-source framework, creating reusable system addons, authentication backends, and dynamic server configurations used by hundreds of developers.
- **Custom Navigation & Pathfinding Runtime:** Wrote a custom C# NavMesh pathfinding engine in Unity enabling dynamic agents to naturally bypass obstacles without disrupting global state logic.
- **Fullstack Enterprise Delivery:** Managed technical execution for over 50 enterprise client projects, setting up code review workflows, git branching standards, and automated regression testing schedules.

FEATURED TECHNICAL INNOVATIONS

- **Mercari Shopping AI-Assistant (Agent Harness):** Built a resilient 3-tier fallback agent discovery engine with MCP-services, dual-LLM provider failovers (Claude to GPT-4o), prompt-injection defense guardrails, interactive clarification, self-correction and persistent session memory.
- **FinTech Data Pipeline & OCR Parser:** Built an automated Python pipeline using OCR and LLMs to parse, validate, and migrate unstructured financial statement records (bank/credit-cards) into Pydantic Schema for direct-import in bookkeeping tools and present a merchant-based spend summary to the user.
- **Custom Unity Firebase C# SDK:** Authored a production-grade C# SDK modernizing an open-source Firebase library, fixing platform build bugs to connect client runtimes directly to Firebase Auth, Firestore, and Cloud Functions.
- **QuickCom Real-Time Engine:** Built a full-stack web scraper using **JavaScript** and **WebSockets** that searches and compares live pricing across grocery delivery platforms in real time.
- **AR Computer Vision POC:** Programmed an augmented reality mobile application using computer vision capable of recognizing 7 to 11 physical playing cards simultaneously from a standard deck to display real-time analysis overlays.

EDUCATION

Bachelor of Science in Information Technology (B.Sc IT)

Amity University, Noida, India